

EYE-GAZE / TEXT-ENTRY EXPERIMENTAL ANALYSIS

A data-driven report of typing speed, accuracy, corrections, keystroke timing, pauses, task difficulty and method-level performance.

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4	1	N/A	35
Rows / Trials	Methods	Participants	Variables

1. Executive Summary

2.24 Mean WPM	75.86% Text Accuracy	79.38 s Mean Time	1.25 Mean Backspaces
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- Average typing speed was **2.24 WPM**, with a range of 1.89–2.85 WPM.
- Final-response text accuracy averaged **75.86%**.
- Mean Levenshtein edit distance was **9.25** characters.
- The user produced an average of **1.25 backspaces per trial**.
- Mean inter-keystroke interval was **4.52 seconds**, providing a direct measure of typing rhythm.

2. Dataset & Variable Inventory

Property	Value
Files loaded	1
Rows	4
Columns	35
Methods	1
Participants	Not detected
Trials	4

Detected columns

Logical Variable	CSV Column
method	method
trial	trial_number
target	stimulus
entered	typed_response
correct	is_correct
start_time	trial_start_time
end_time	trial_end_time
duration	typing_duration_sec
char_count	char_count
word_count	word_count
spaces	spacebar_count
backspaces	backspace_count
timestamps	letter_timestamps
wpm	__wpm

Logical Variable	CSV Column
cps	__cps
edit_distance	__edit_distance
normalized_error	__normalized_error_pct
final_correct	__final_response_correct
text_accuracy	__text_accuracy
correction_rate	__correction_rate
event_count	__event_count
iki_mean	__iki_mean
iki_median	__iki_median
iki_sd	__iki_sd
iki_min	__iki_min
iki_max	__iki_max
pause_count_3s	__pause_count_3s
pause_count_5s	__pause_count_5s
longest_pause	__longest_pause
first_event	__first_event
last_event	__last_event
backspace_events	__backspace_events
space_events	__space_events

3. Core Typing Performance

Metric	Mean	SD	Median	Min	Max	N
WPM	2.24	0.43	2.10	1.89	2.85	4
CPS	0.19	0.04	0.17	0.16	0.24	4
Completion time (s)	79.38	85.90	45.25	21.02	206.01	4
Characters	14.25	15.35	7.50	5.00	37.00	4
Words	3.00	3.37	1.50	1.00	8.00	4
Backspaces	1.25	1.89	0.50	0.00	4.00	4
Spaces	2.00	3.37	0.50	0.00	7.00	4
Text accuracy (%)	75.86	37.16	90.91	21.62	100.00	4
Edit distance	9.25	13.70	4.00	0.00	29.00	4
Normalized error (%)	24.14	37.16	9.09	0.00	78.38	4
Correction rate (%)	7.70	9.65	5.41	0.00	20.00	4
Mean IKI (s)	4.52	0.68	4.33	3.95	5.47	4
Median IKI (s)	4.57	0.94	4.20	3.95	5.94	4
Longest pause (s)	7.33	1.79	7.17	5.71	9.27	4

Metric definitions

WPM = (characters typed / 5) / duration in minutes.

CPS = characters typed / duration in seconds.

Edit distance = minimum insertions, deletions and substitutions needed to transform the entered response into the target.

Correction rate = backspaces / final character count × 100.

IKI = time between consecutive recorded keystroke events.

4. Trial-Level Analysis

Trial	Method	Target	Entered	Time	WPM	Accuracy	BS	Edit
1	overt	smile	smile	21.02	2.85	100.00%	0	0
2	overt	light	light	31.74	1.89	100.00%	1	0
3	overt	The quick brown fox jumps the quick fox jumps over the	the quick fox jumps over the	206.01	2.16	81.82%	4	8
4	overt	Waltz, bad nymph, for quick	waltz jinx	58.76	2.04	21.62%	0	29

5. Keystroke-Level Analysis

Trial	Events	Mean IKI	Median IKI	IKI SD	3s Pauses	5s Pauses	Longest Pause
1	5	4.10	3.95	1.45	2	1	5.71
2	7	3.95	3.95	1.27	5	1	5.92
3	45	4.55	4.44	1.73	36	15	9.27
4	10	5.47	5.94	1.99	7	5	8.43

The timestamp stream permits analysis of typing rhythm rather than only final trial performance. Long inter-keystroke intervals can indicate hesitation, search time, correction, or task difficulty. They should not automatically be interpreted as cognitive hesitation without additional evidence.

6. Error & Correction Behavior

Total backspaces across the dataset: **5**. Mean per trial: **1.25**. Maximum in a trial: **4**.

Mean final-response edit distance: **9.25**. This is preferable to relying only on an `is_correct` flag because it quantifies how far the final response is from the target.

The CSV's `is_correct` field reports **50.0%** correct trials. The report separately calculates final-response correctness from target and entered text so inconsistent correctness flags can be identified.

7. Method Comparison

Only one method was detected in the current data, so a method comparison cannot be performed. Multiple methods such as Traditional Keyboard and Eye-Gaze Keyboard are required.

8. Task / Phrase Difficulty

Trial	Target Length	Words	Spaces	Punctuation	Time	WPM
1	5	1	0	0	21.02	2.85
2	5	1	0	0	31.74	1.89
3	44	9	8	1	206.01	2.16
4	37	7	6	3	58.76	2.04

Phrase length, spaces and punctuation provide task-complexity descriptors. Completion time should therefore be interpreted alongside WPM/CPS rather than used alone to compare phrases of different lengths.

9. Relationship Analysis

Relationship	Pearson r	p-value	N
WPM vs Accuracy	0.349	0.6507	4
WPM vs Duration	-0.235	0.7648	4
Duration vs Characters	0.998	0.0016	4
Backspaces vs Edit Distance	-0.183	0.8168	4
Mean IKI vs WPM	-0.284	0.7158	4
Longest Pause vs WPM	-0.386	0.6137	4

10. Gaze & Head-Pose Availability

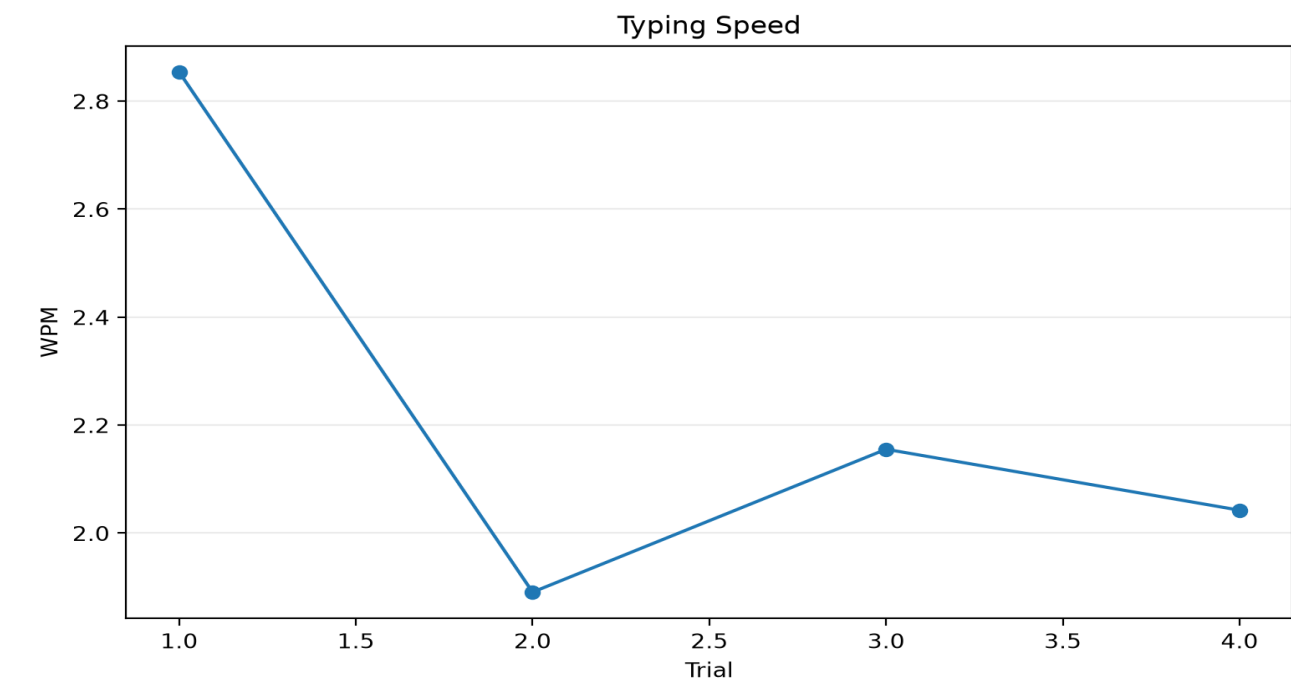
No raw gaze X/Y coordinate stream was detected in this CSV. Therefore gaze trajectory, gaze jitter and EMA smoothing effectiveness cannot be measured from this file.

No yaw/pitch head-pose stream was detected.

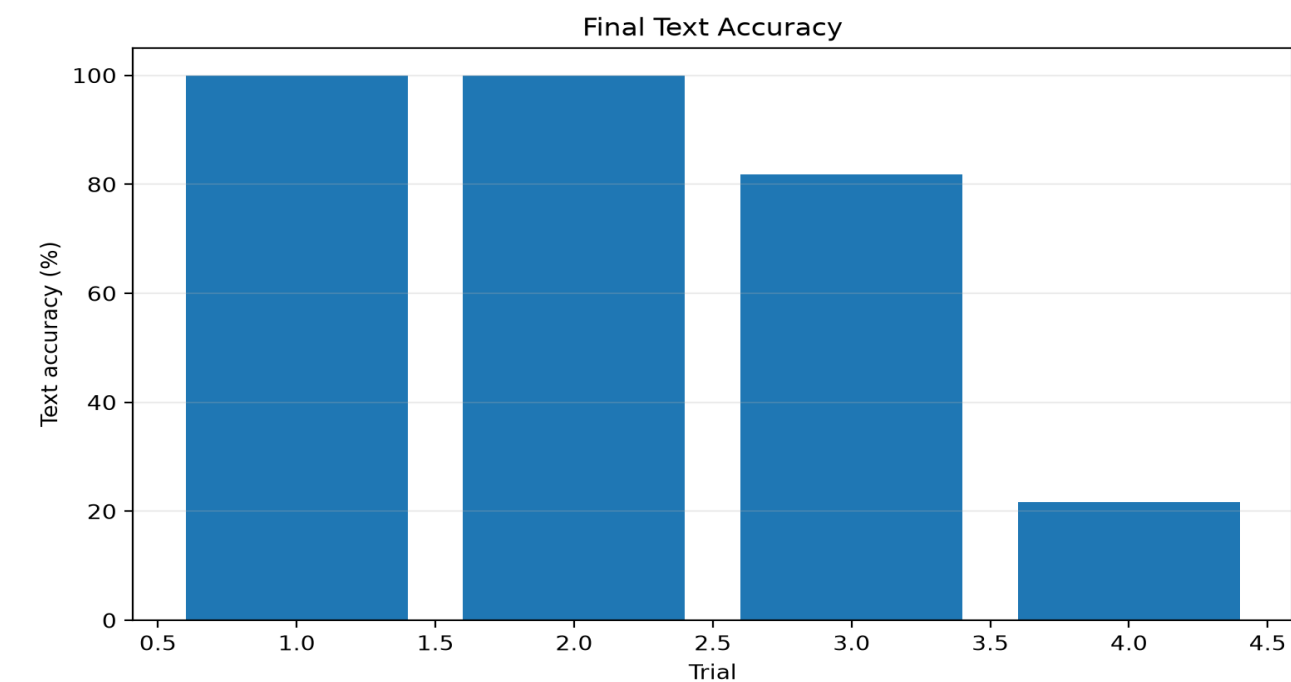
No dwell-time field was detected, so dwell-to-click performance cannot be evaluated from this dataset.

11. Visual Analysis

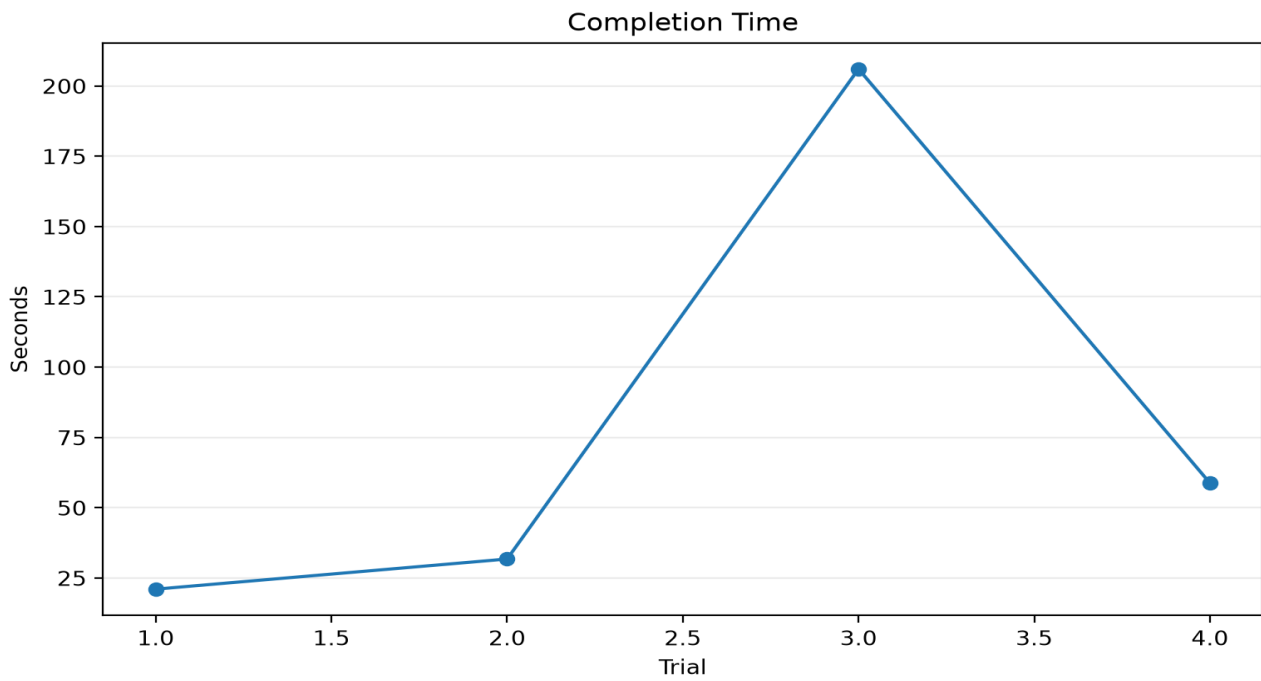
Typing Speed



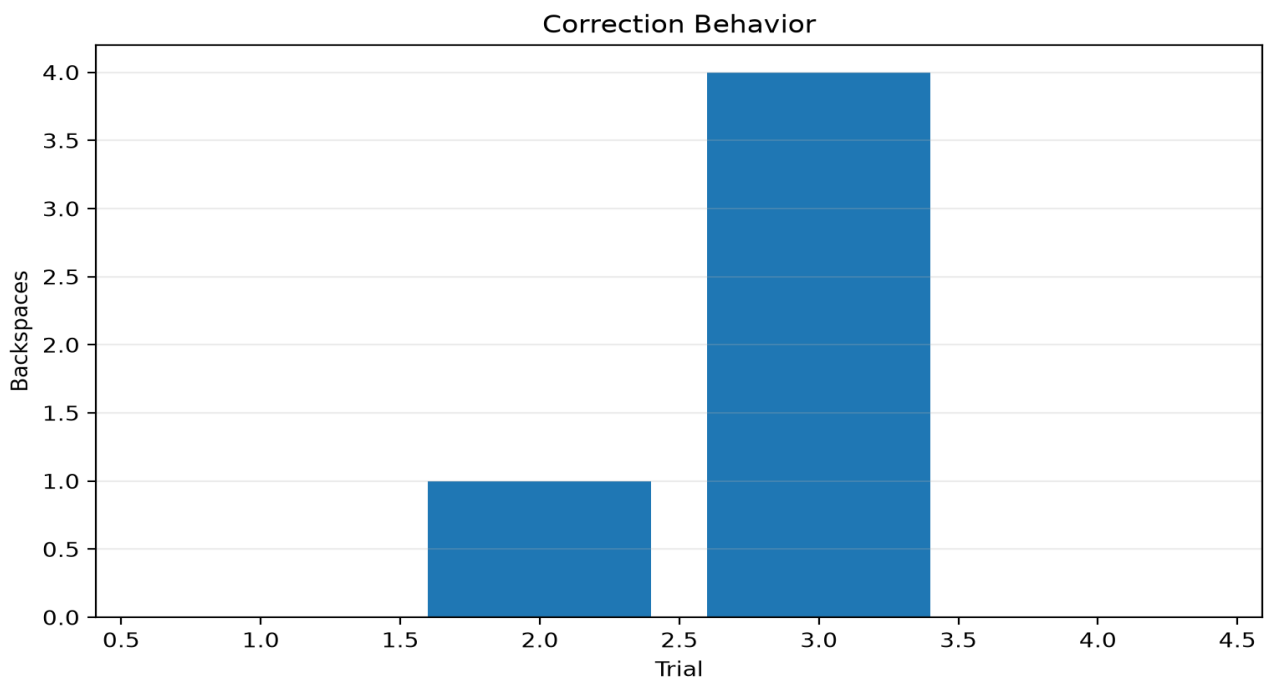
Final Text Accuracy



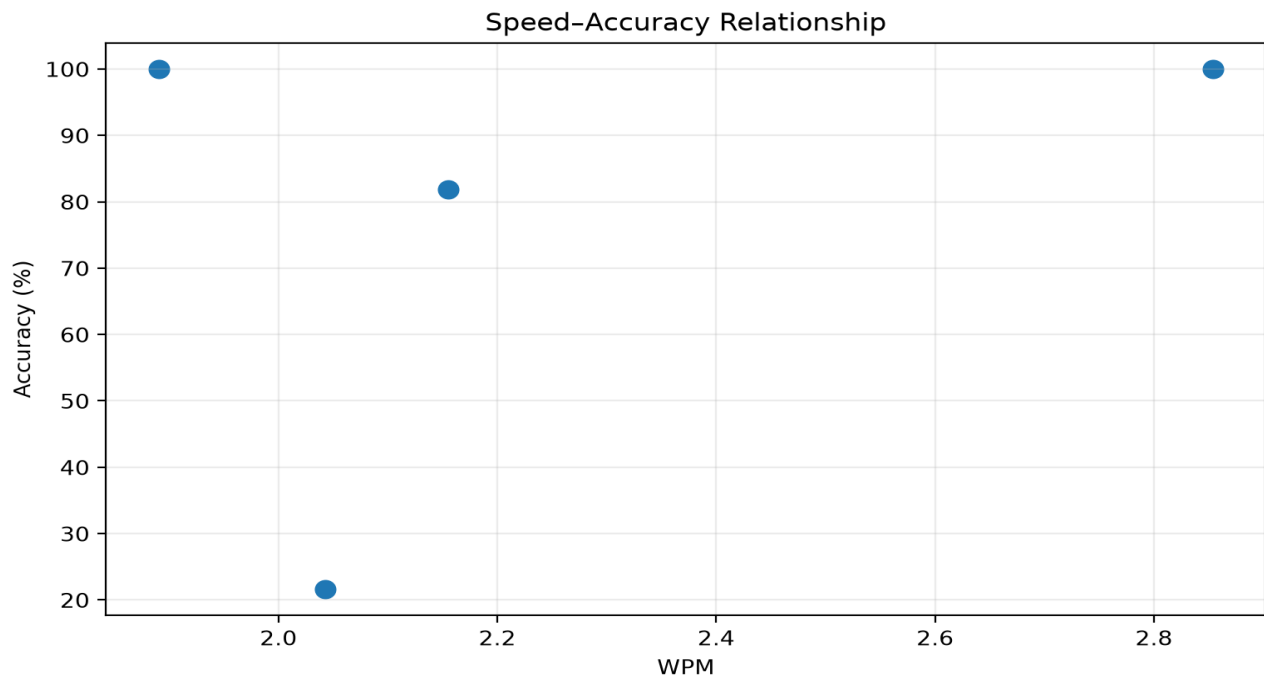
Completion Time



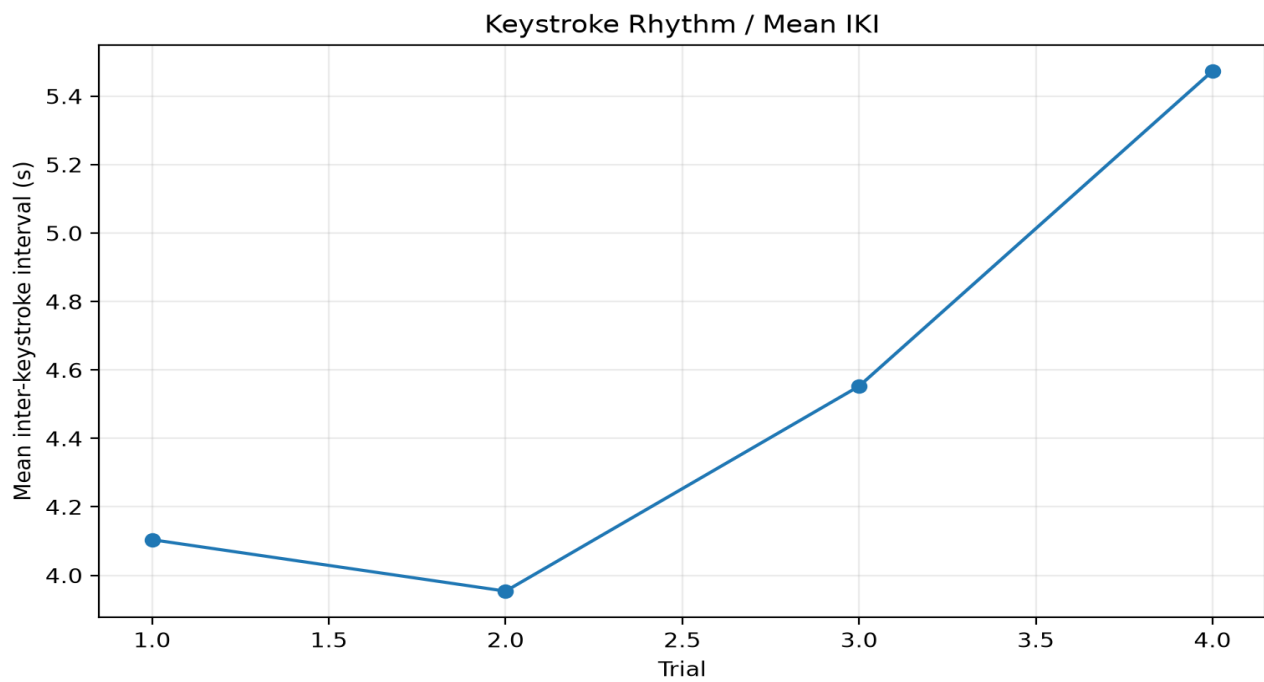
Correction Behavior



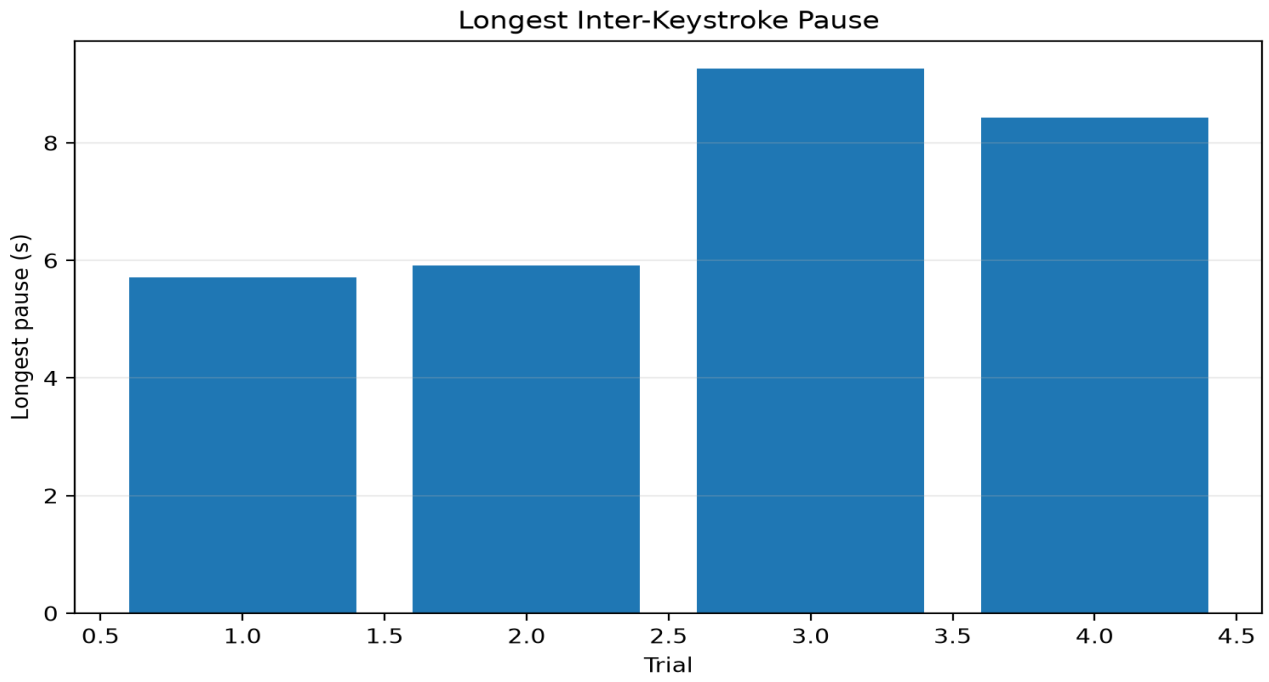
Speed–Accuracy Relationship



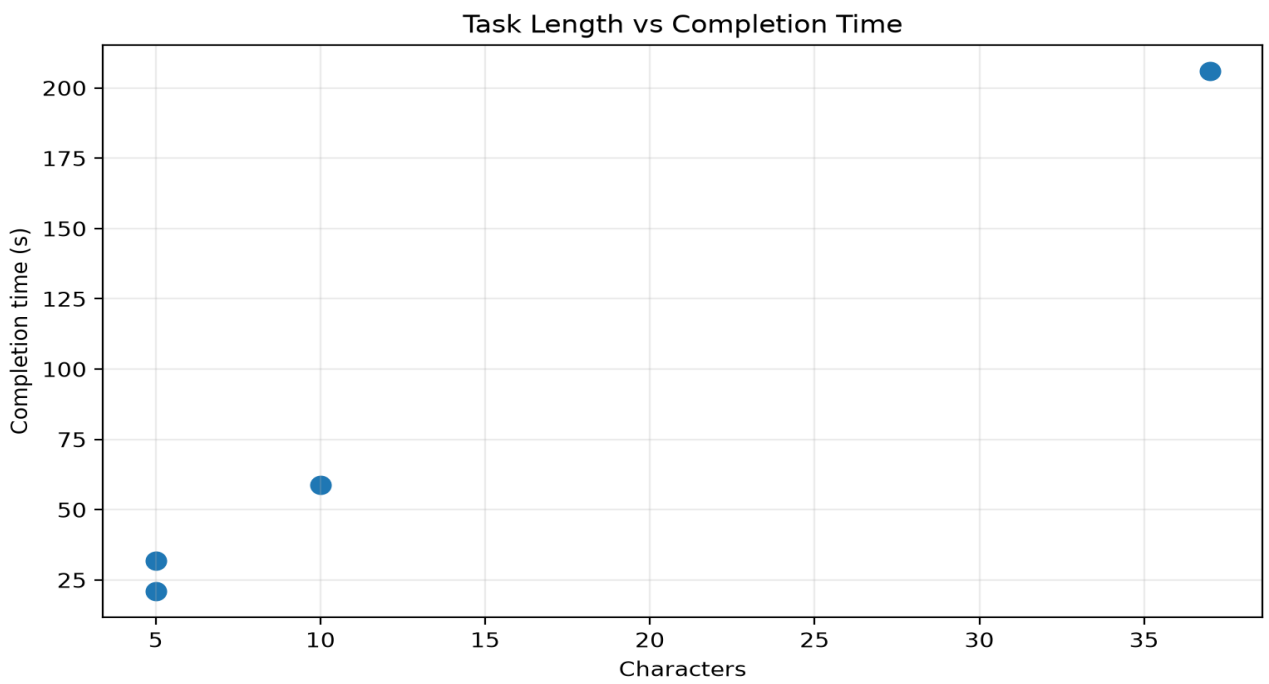
Keystroke Rhythm



Pause Analysis



Task Length vs Completion Time



12. Data Quality & Research Notes

Variable	CSV Column	Missing
method	method	0.0%
trial	trial_number	0.0%
target	stimulus	0.0%
entered	typed_response	0.0%
correct	is_correct	0.0%
start_time	trial_start_time	0.0%
end_time	trial_end_time	0.0%
duration	typing_duration_sec	0.0%
char_count	char_count	0.0%
word_count	word_count	0.0%
spaces	spacebar_count	0.0%
backspaces	backspace_count	0.0%
timestamps	letter_timestamps	0.0%
wpm	__wpm	0.0%
cps	__cps	0.0%
edit_distance	__edit_distance	0.0%
normalized_error	__normalized_error_pct	0.0%
final_correct	__final_response_correct	0.0%
text_accuracy	__text_accuracy	0.0%
correction_rate	__correction_rate	0.0%
event_count	__event_count	0.0%
iki_mean	__iki_mean	0.0%
iki_median	__iki_median	0.0%
iki_sd	__iki_sd	0.0%
iki_min	__iki_min	0.0%
iki_max	__iki_max	0.0%
pause_count_3s	__pause_count_3s	0.0%
pause_count_5s	__pause_count_5s	0.0%
longest_pause	__longest_pause	0.0%
first_event	__first_event	0.0%
last_event	__last_event	0.0%

Variable	CSV Column	Missing
backspace_events	__backspace_events	0.0%
space_events	__space_events	0.0%

Important observations

- Only 4 rows are available. This is suitable for validating the analysis pipeline but is too small for strong statistical/generalization claims.
- Only one input method is present. Comparative claims require at least two conditions.

Research interpretation: descriptive differences are not automatically evidence of superiority. For a thesis-quality comparison, collect repeated trials across multiple participants under matched task conditions and analyze participant-level paired observations.

Appendix A — Source Files

experiment_overt_brittick_20260907_162721.csv

Appendix B — Processed Dataset

The output folder contains combined_processed_data.csv. Derived columns are prefixed with ___ so the original CSV fields can be distinguished from calculated metrics.